Data Scientist - Machine Learning - Group Exam

Markus Lindell

May 2026

The goal of this project was to build a machine learning model capable of identifying potentially suspicious marketplace listings and helping a stakeholder prioritize which listings should be manually reviewed. The project began with exploratory data analysis where the dataset was examined in terms of data types, missing values, and target distribution. One important finding was that the dataset was highly imbalanced, with significantly more normal listings than suspicious ones.

To avoid data leakage, the data was split into training and test data before preprocessing was applied. Missing values were handled by creating missing-value indicators and performing imputation within a preprocessing pipeline together with scaling and one-hot encoding of categorical variables. Several models were trained and compared, including a baseline model, Logistic Regression, and a Random Forest classifier. Since the dataset was imbalanced, PR-AUC was selected as the primary evaluation metric rather than accuracy.

After comparing model performance, Logistic Regression was selected as the final model because it achieved the strongest PR-AUC performance while maintaining stable results across validation experiments. Hyperparameter tuning was performed using GridSearchCV, although only small improvements were observed. The final model was evaluated on previously unseen test data and achieved a PR-AUC score of approximately 0.27. Since only around 10% of listings were suspicious, a random classifier would be expected to achieve a PR-AUC close to 0.10, meaning the model performed substantially better than random ranking.

Finally, the trained model was applied to new data where each listing received a risk score and a prioritization ranking. A threshold of 0.60 was selected so that high-risk listings would be flagged for manual review rather than automatically removed. The project showed that machine learning can improve efficiency in the review process while also demonstrating that human judgment remains important because the model estimates risk rather than determining with certainty whether a listing is fraudulent.